

Visualization, Intro

MGT 100 Week 1

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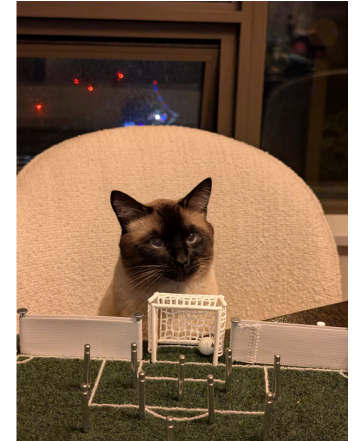
[Daniel Yavorsky](#)

SVP-Analytics

GBK Collective

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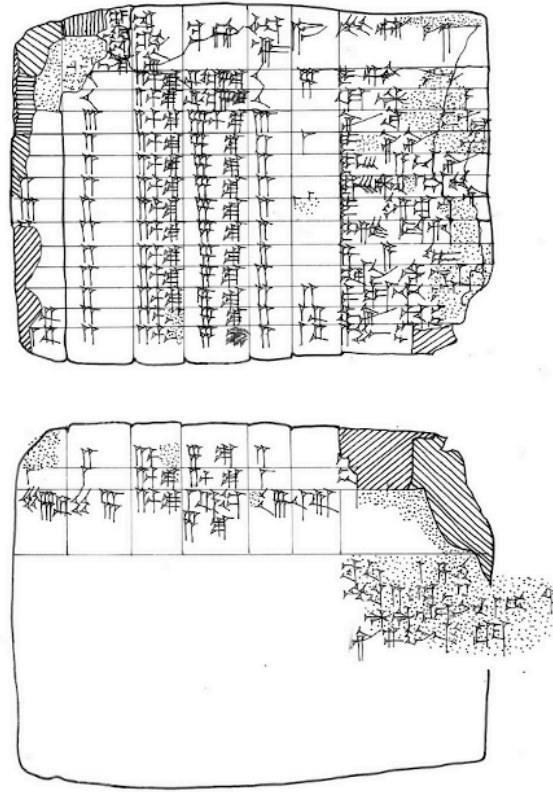
Dog==Draymond || Dray || Click Clack || Major Jealous
Cat==Luka || Liquid || Hunter || Octopus

Data Visualizations (Viz)

- “The greatest value of a picture is when it forces us to notice what we never expected to see.”
 - Boxplot Inventor John Tukey
 - Great visualizations raise new questions

Tukey (1977)

Data Viz are older than 0



This ancient Mesopotamian clay tablet is estimated to be 3,500 years old. Data visualization is older than English and older than zero. Are those blanks zeros or missing values?

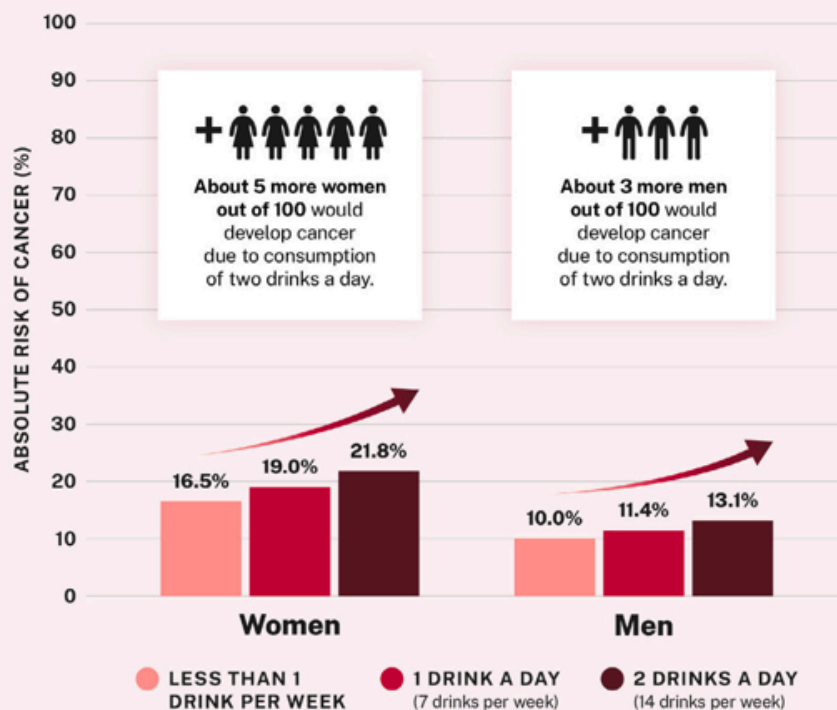
[source](#)

Translated

A	B	C	D	E	F	G
[15]	10	2½ s.	½ m. 5 s.	10	5	Igigi
[10]	10	2½ s.	½ m. 5 s.	10		Awat-Šamaš
[3]	3	2½ s.	7½ s.	3		Nidittum
[3]	2	2½ s.	5 s.	2	1	The sons of Ili-iddinam
[2]	2	2½ s.	5 s.	2		The sons of Ahuni
[2]	2	2½ s.	5 s.	2		The sons of Sin-šamuh
3	2	2½ s.	5 s.	2	1	The son of Imgurum
2	2	2½ s.	5 s.	2		Isi-dare
[2]	2	2½ s.	5 s.	2		Zayyalum
[2]	2	2½ s.	5 s.	2		Silli-Ištar
[4]	2	2½ s.	5 s.	2	2	Paqatiya
[3] ½	2	2½ s.	5 s.	2	1½	Perhum
[3] ½	2	2½ s.	5 s.	2	1½	Abum-ili
[2]	2	2½ s.	5 s.	2		[Sin-ašareda]
[1]	1	2½ s.	[2½ s.]	1		Aplum
57 [½]	46	2½ s.	1½ m. 5 s.	46	11½	
Tašritum (Month VI), [day...]						
Year: the third after Isin was destroyed by the great weapons of An, Enlil and Enki						
A = "Troops, available assets" B = "Those who were sent <i>from</i> Sin's dyke" C = "Earth, the (daily) work assignment of 1 man" D = "Its earth" E = "Check made" F = "Arrears" G = "Its name"						

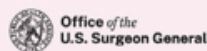
This attempted translation shows what appears to be accounting for construction labor. Columns B and E look like duplicates, and A appears to sum B and F. Are those blanks zeros or missing data?

Higher alcohol consumption increases alcohol-related cancer risk in women and men



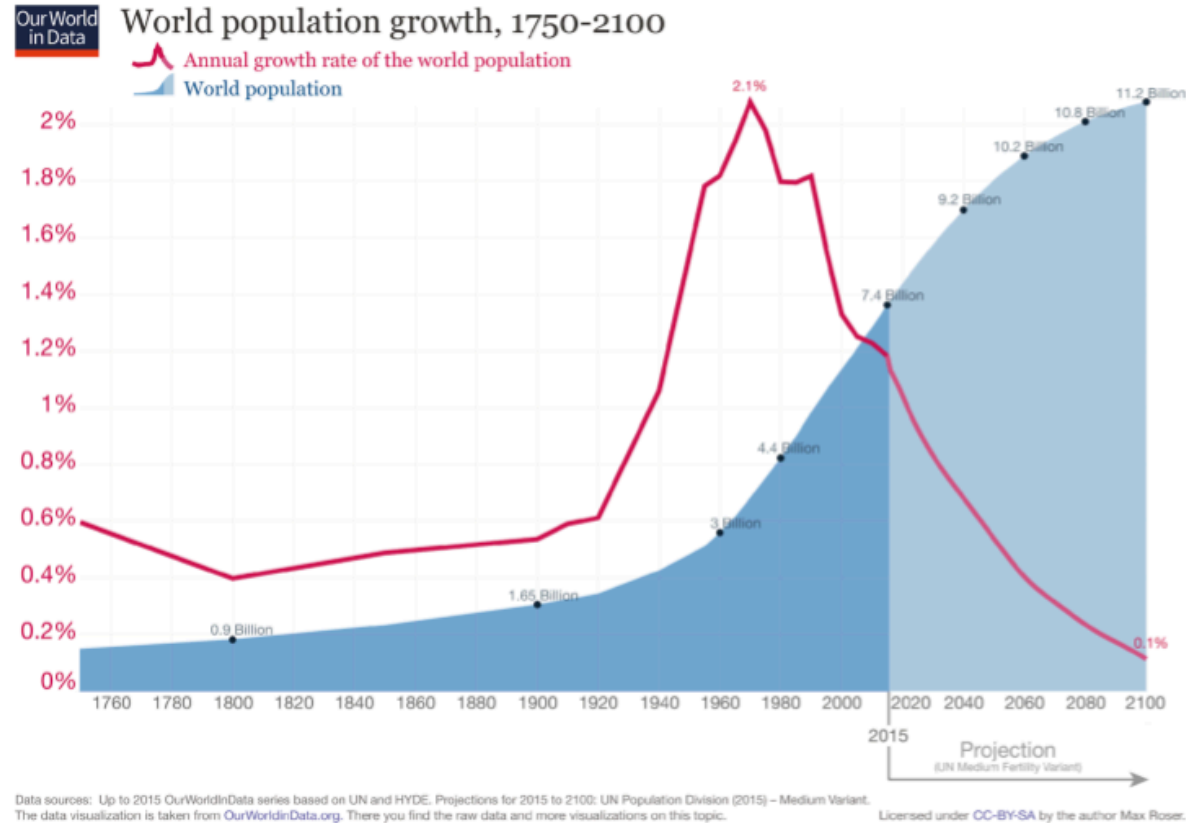
This graph represents the cumulative absolute risk of alcohol-related cancer in women and men over the lifespan by age 80. Alcohol-related cancer includes breast, colorectum, esophagus, liver, mouth, throat, and voice box cancers.

Source: Calculated with data from Sarich, P., Canfell, K., Egger, S., Banks, E., Joshy, G., Grogan, P., & Weber, M. F. (2021). Alcohol consumption, drinking patterns and cancer incidence in an Australian cohort of 226,162 participants aged 45 years and over. *British journal of cancer*, 124(2), 513–523. <https://doi.org/10.1038/s41416-020-01101-2>

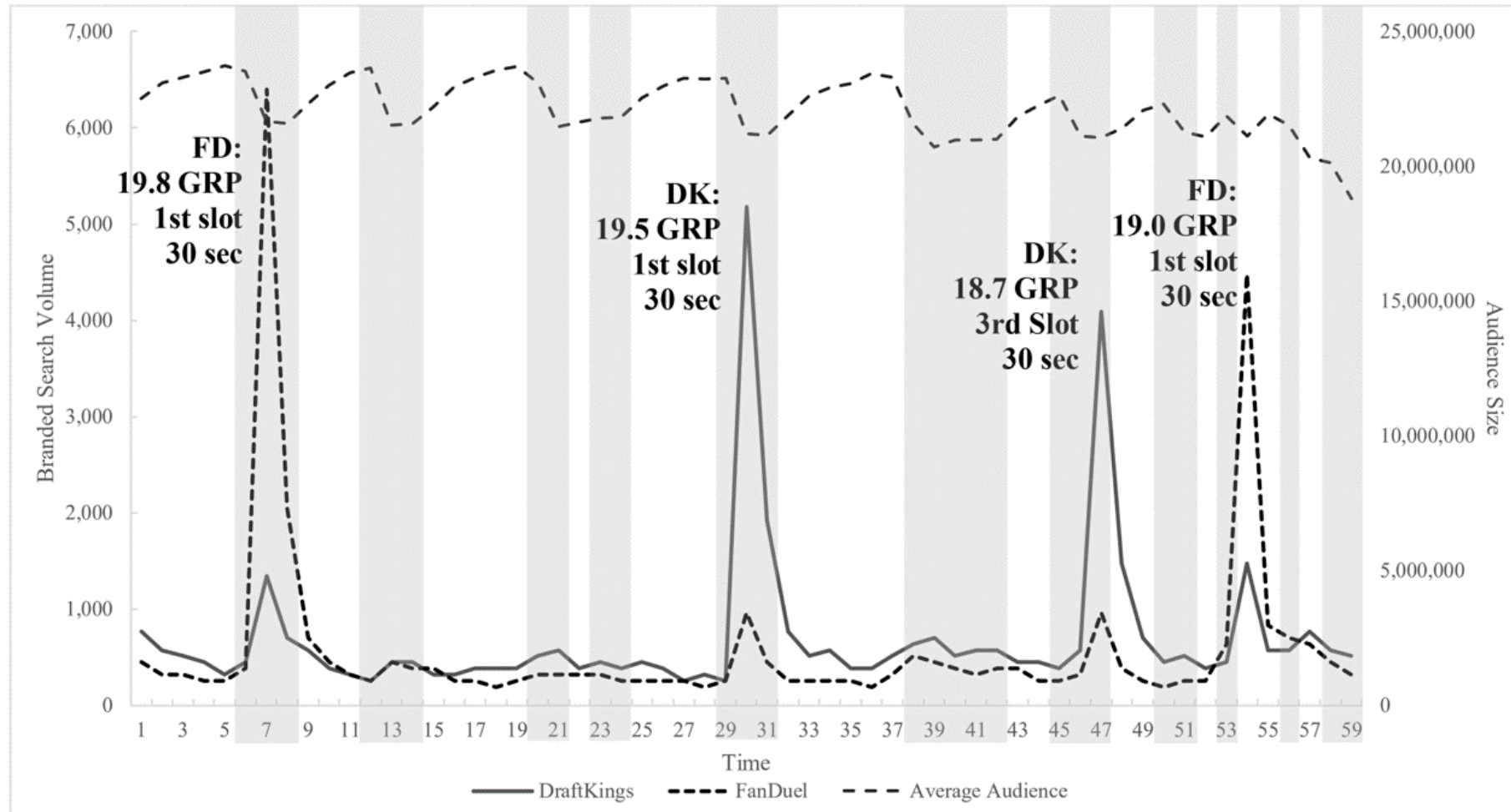


This visualization shows absolute increase in cancer risk and offers information for different audiences, with a pleasing presentation despite a grave topic.

Does the graph show how much a person's cancer risk changes when they change their drinking behavior? (I.e., a "causal effect")

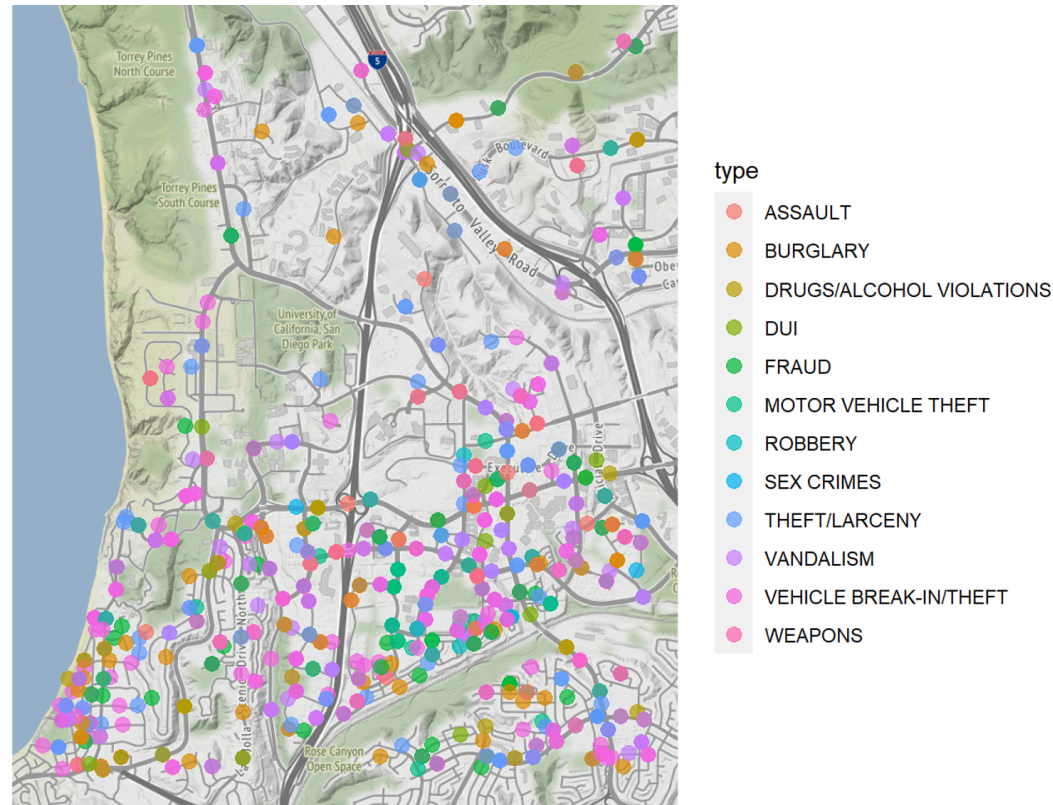


The world population curve has a sigmoid shape, also known as an S curve, in which growth increases and later decreases. A pop science book, *Population Bomb* by Ehrlich (1968), forecast mass starvation, since population was increasing exponentially but food production was increasing linearly.



Google searches for DraftKings, FanDuel during an NFL Game, 9-10 P.M. EST. Commercial minutes are shaded. What do you see?

SDPD Crime Reports Near UCSD



When is data zero vs. missing? Provenance is key. What kind of decisions could this inform?

Hansen

Data *Provenance*

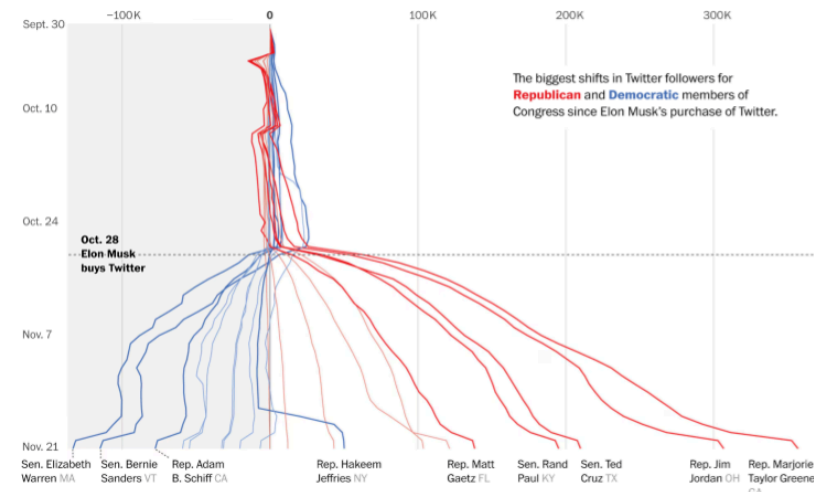
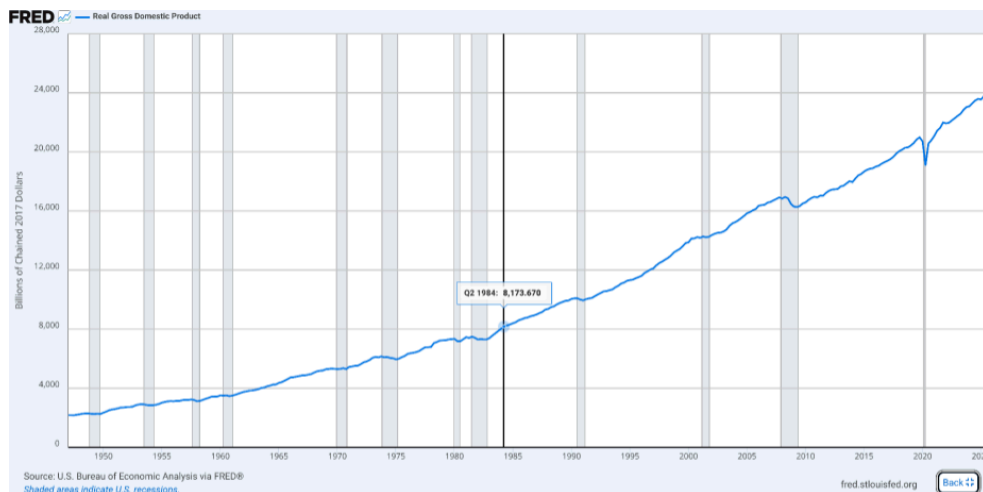
- Describes people, entities, and activities involved in producing, compiling, transforming & sharing data
 - Verifiability requires source data & cleaning scripts
- Crucial in determining quality, reliability, trustworthiness
 - Investigating provenance requires and deepens domain expertise
 - Typically turns up unexpected information, and sometimes errors
- When can you trust provenance information?
 - Best to treat provenance as hypotheses to be verified in the data
 - Usually, provenance descriptions are missing or imperfect
 - Financial variables are most likely to be accurate due to auditing requirements
 - Considerations: Consent; Privacy; Missingness; Permissible uses
 - Sometimes, provenance descriptions are marketing documents

GIGO: Garbage In, Garbage Out. Don't analyze noise.

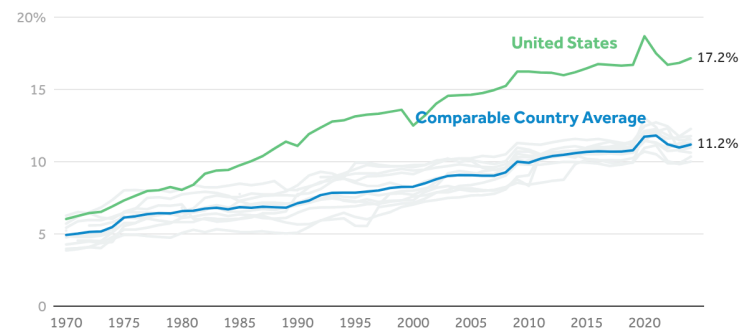
Jungco (2025)

Viz can build trust, understanding

- Eyeballs can interpret pictures quickly
 - Human brains are great at interpreting visual patterns, predictions
 - Can detect unexpected errors
- Understandable to non-specialists, including customers and managers
 - Viz choices enable narrative understanding, another common brain function
- Viz are *lingua franca* across disciplines
 - Easily replicated -> more easily trusted
 - Trust can be a major issue in some corporate cultures
- Viz succeed when they raise deeper questions
 - Asking the next question indicates acceptance; facts prompt explanations
 - “I wonder why that is...” or “Maybe that’s because...”
 - Viz usually won’t settle major debates, but viz is the first step



Health consumption expenditures as percent of GDP, 1970-2024



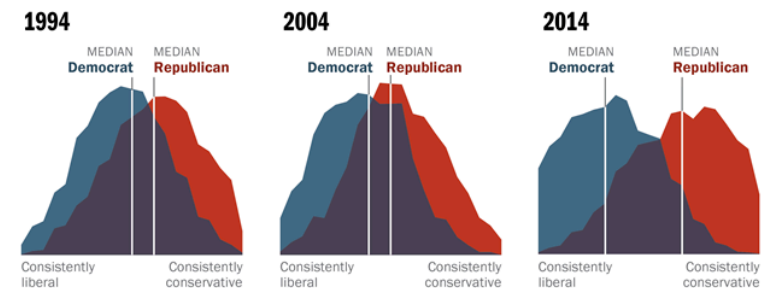
Notes: Health consumption does not include investments in structures, equipment, or research. See Methods section of How does health spending in the U.S. compare to other countries? for details.

Source: KFF analysis of OECD for comparable countries, and CMS National Health Expenditure historical data for the United States.

Peterson-KFF
Health System Tracker

Democrats and Republicans More Ideologically Divided than in the Past

Distribution of Democrats and Republicans on a 10-item scale of political values

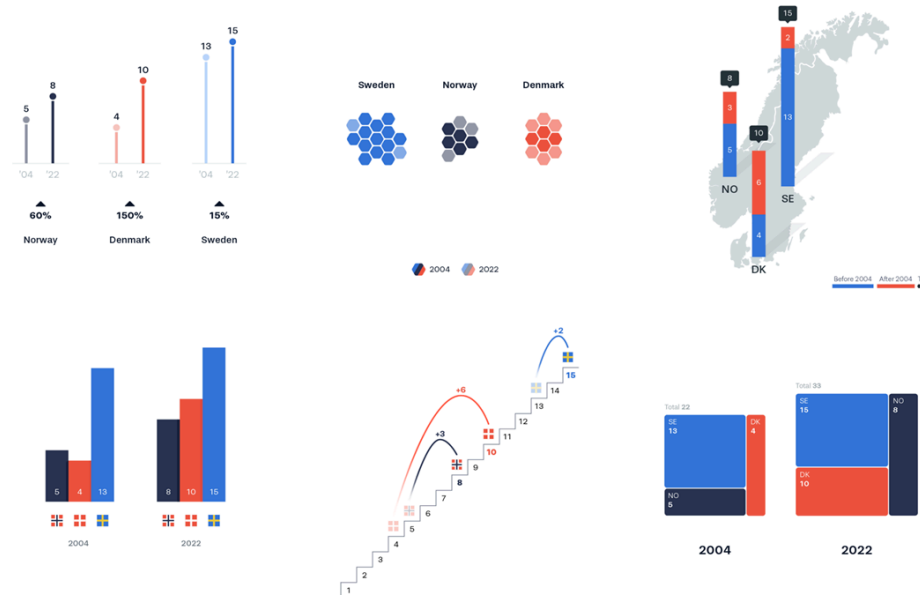


Source: 2014 Political Polarization in the American Public
Notes: Ideological consistency based on a scale of 10 political values questions (see Appendix A). The blue area in this chart represents the ideological distribution of Democrats; the red area of Republicans. The overlap of these two distributions is shaded purple. Republicans include Republican-leaning independents; Democrats include Democratic-leaning independents (see Appendix B).

PEW RESEARCH CENTER

“Graphics journalists urge that each chart should make exactly one point – and it should be obvious how to read it. Often charts say (1) number goes up/down recently, (2) number goes up/down when some event occurred, (3) one set of lines diverges from another set of lines (or, one line is an outlier compared to the rest), (4) the distribution is bimodal.” –Jeremy Merrill, WaPo. “Scientific charts are often the opposite of this – they have four variables, six symbols and are explained two pages away.”

Number of World Heritage Sites			
	Norway	Denmark	Sweden
2004	5	4	13
2022	8	10	15



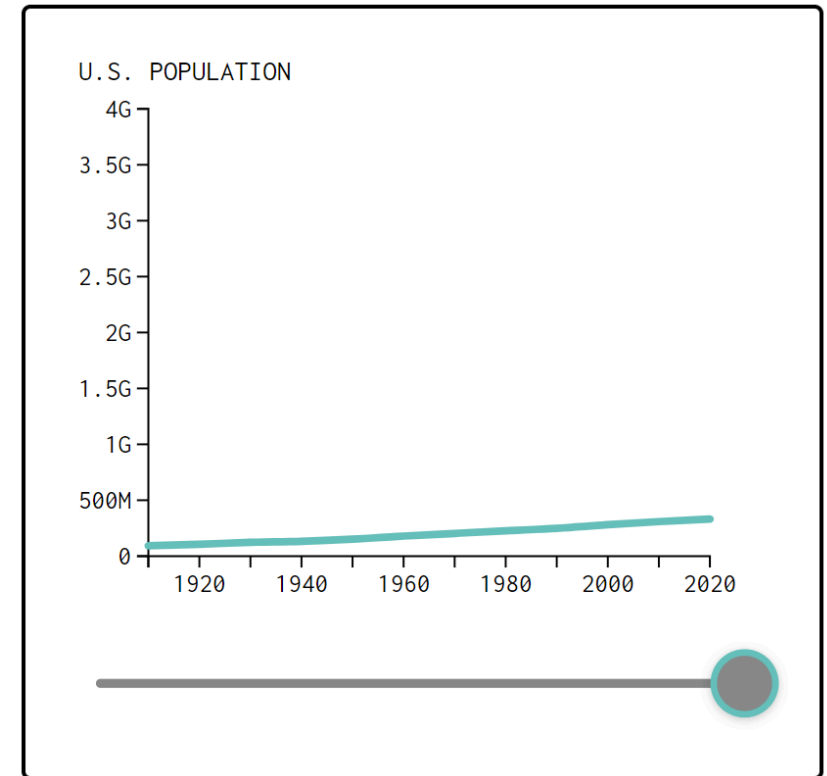
The same dataset can be represented many different ways, each emphasizing different comparisons. Here is a very simple dataset; what comparison does each graphic communicate? How would you summarize each graphic in a sentence?

Honest Viz

1. Usually start axes from 0, choose scales judiciously
2. Show all relevant data, label accurately
3. Discretize and smooth judiciously
4. Show uncertainty when relevant
5. Cite data sources

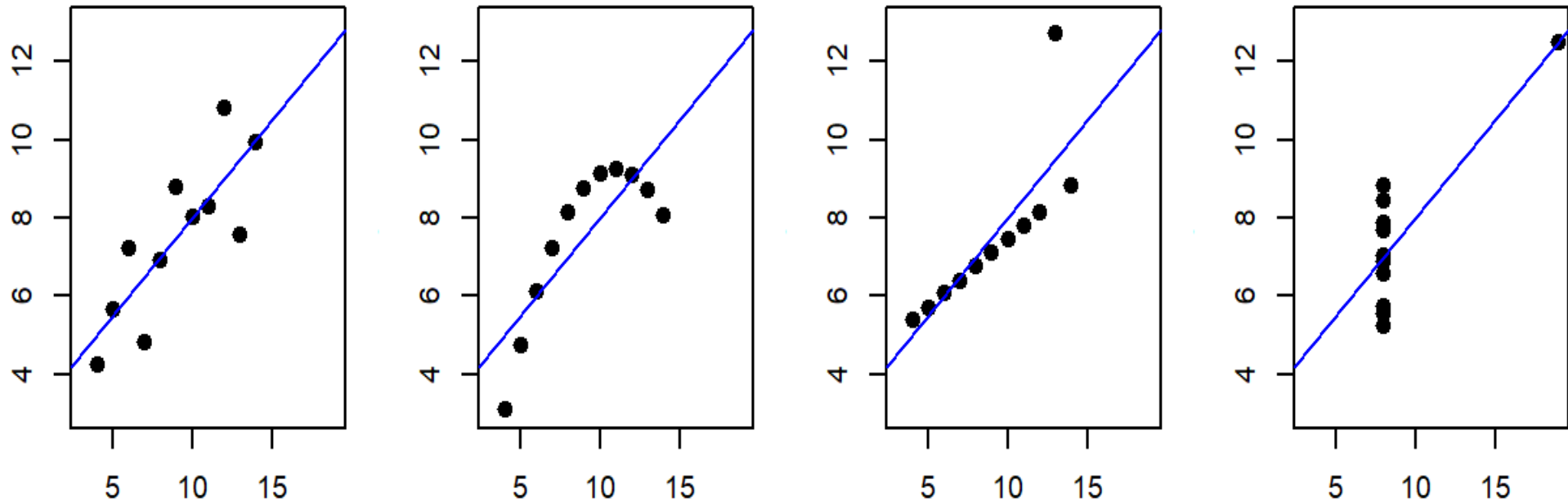
Misleading visualizations can indicate bad faith. You want to avoid mistaken interpretations of your work, and identify misleading impressions created by others.

[Go deeper](#)



Visualize data before you model anything

- 4 Datasets, Same $\hat{\beta}^{OLS}$, i.e. same $\frac{x'x}{x'y}$



What does each picture tell us? This is why we shouldn't just use regression to summarize data

Customer Analytics

Customer

- Receives good or service in exchange for payment (money, time, attention)
- Has agency: Can say “no”
- “The purpose of business is to create and keep a customer.”
-Drucker
- “There is only one boss. The customer. And he can fire everybody in the company from the chairman on down, simply by spending his money somewhere else.” -Walton
- “In the long term, there’s never any misalignment between customer interests and shareholder interests.” -Bezos

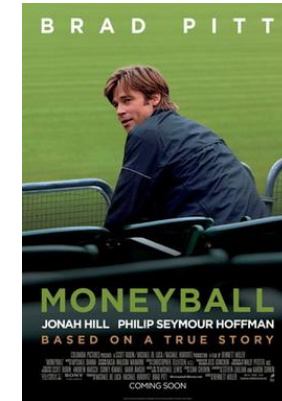
related Consumer, Client



Analytics

- Using data to improve decisions
- Popularized by Charles Taylor in 1910s
- Further popularized by *Moneyball* (2011)
- Measurement, Heuristics, Graphics, Models, Predictions, Automation, Optimization, Personalization, ...
- Can be deceptively difficult

related Expert systems, business intelligence, data science, AI.
Terms change frequently.



First Law of Customer Analytics

- No Customers, No Business
- No Customers ->
No Revenue ->
No Profit ->
No Business
QED

Which owners or employees in the business can afford to ignore customers?

Second Law of Customer Analytics

- More Customers, More Profits
- More Customers ->
More Revenue ->
More Profit
QED
 - These are empirical tendencies, not logical necessities
 - Individual customers can be unprofitable if $(\text{price} - \text{cost}) < 0$
- Marketing Objective: Maximize Long-term Profits
 - Long-term focus mostly aligns our interest with customers
 - Goal is not only to acquire customers; also, keep and develop them
 - Sidesteps or reduces most ethical dilemmas
 - Short-run objective may profit prior to liability; I won't teach

Example of Great Marketing: Netflix

- Consumer surplus
 - Suppose customer pays \$20/month, watches 60 hours: \$0.33 per entertainment hour, or \$0.13/hour with ads, or less with shared accounts
 - A la carte rentals are more like \$1-2+/hour
 - Non-video entertainment tends to be much more expensive
 - Social benefits may accrue from shared viewing
 - Other streaming services may be competitive
- Producer surplus
 - NFLX spent \$13B on content in 2023, 230 million subscribers, about \$4.70/user/month
 - Cost structure: High fixed, low marginal
 - Ads likely earning around \$4-5 per recipient/month
 - Net profit margin about 20-22% in a competitive category

Other EDLP (vs High-Low) businesses: Costco, Trader Joes, Walmart. Common tension between short-term management and long-term strategies.

How can we use customer data?

- Businesses have 4, and only 4, ways to make money:
Acquire, develop, retain and “fire” customers
 - This is called Customer Relationship Management (CRM): week 9
- Marketing mix (“4 P’s”): Improve product offerings, prices, promotion, distribution
- Incorporate customer heterogeneity for targeting, personalization, recommendations, product development...
- Privacy and security, e.g. misuse, theft, regulatory compliance

Firing customers is typically indirect, such as withdrawing preferred products or declining to encourage further purchasing

Example: Nielsen

CONSUMER PANEL DATA

- The Consumer Panel Data include longitudinal data beginning in 2004. These data track a panel of 40,000–60,000 US households and their purchases of fast-moving consumer goods from a wide range of retail outlets across all US markets.

RETAIL SCANNER DATA

- Retail Scanner Data consist of weekly pricing, volume, and in-store marketing info generated by point-of-sale systems from 90+ participating retail chains across all US markets. Data begin in 2006.

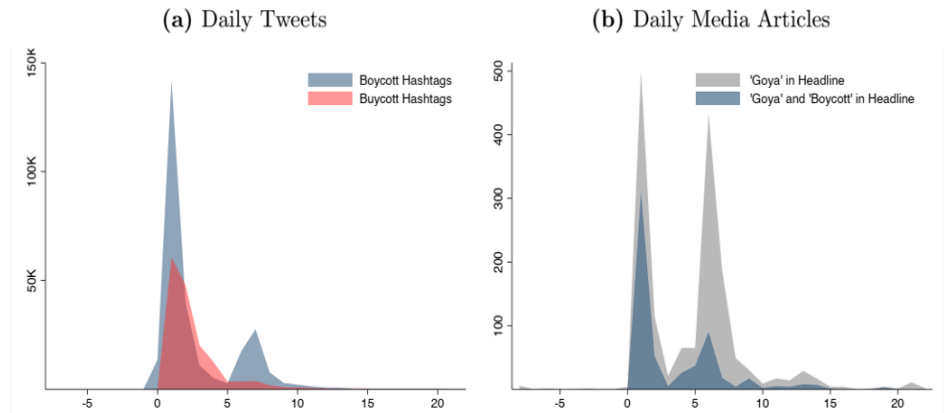
Consumer panel and retail scanner data are foundational to customer analytics in consumer packaged goods. What business questions could these data help answer?

Figure 1: Images of Tweets Triggering Goya Boycott and Buycott



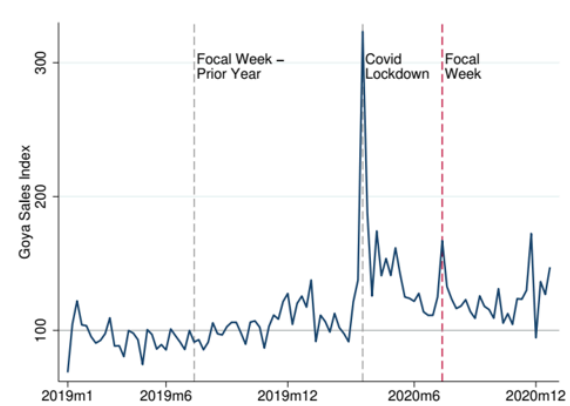
On July 9, 2020, the CEO of Goya praised President Trump during a White House meeting, generating calls for a consumer boycott.

Figure 2: Goya Related Daily Tweets and Media Coverage in July 2020



Notes: Day 0 is July 9th, 2020. Panel (a) depicts the number of daily tweets in July 2020 that use boycott- or buycott-related hashtags. Web Appendix A.3 details the classification procedure and Table A1 lists the hashtags. Panel (b) plots the number of daily media articles that include 'Goya' and 'Boycott' in the headline. Only 18 articles included buycott related keywords so they are not depicted. In both panels, the second spike is triggered by President Trump and his daughter's tweets in support of Goya.

Figure 3: Total Weekly Spending Index on Goya Products

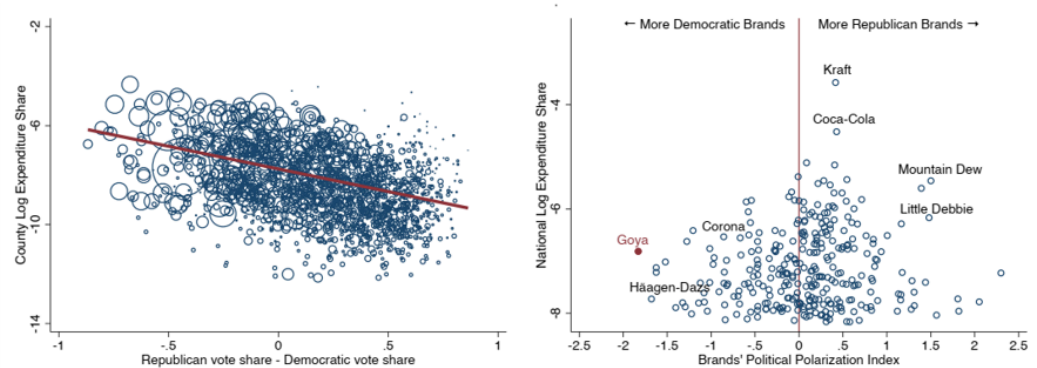


Notes: This figure depicts weekly spending index on Goya products from 2019-2020. The pre-pandemic sales average is normalized to 100. The red vertical dashed line indicates the focal week of the scandal (July 9-15). The spike and subsequent dip in November 2020 is due to bunching in purchases around the Thanksgiving holiday. Please see Appendix B.8 for a replication of the sales trend using Nielsen RMS data.

Despite calls for a boycott, total sales rose, mostly because Republican areas started buying Goya. Without customer data, we would be shooting in the dark.

Figure 5: Polarization in Brand Popularity: Goya vs Other Top Brands

(a) Goya Market Share and Vote Margin (b) Polarization of Goya and Other Top Brands



Notes: Panel (a) depicts the relationship between Goya's log expenditure share in a given county in 2019 and the vote margin in that county. The size of the bubble is proportional to the number of panelists in a given county. Panel (b) plots the estimated political polarization index for each of the top 301 CPG brands. The index quantifies the extent to which the market share of a brand varies with political preferences. Goya's national expenditure share is 0.11% (the log of its national expenditure share is -6.81). See Web Appendix A.1 for more detail.

4 TYPES OF ANALYTICS

Correlational

Looking Back

1. Descriptive

What's happening with my business?

- Comprehensive, accurate, and live data
- Effective Visualization

Causal

2. Diagnostic

Why is it happening?

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

Looking Ahead

3. Predictive

What's likely to happen?

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

4. Prescriptive

What do I need to do?

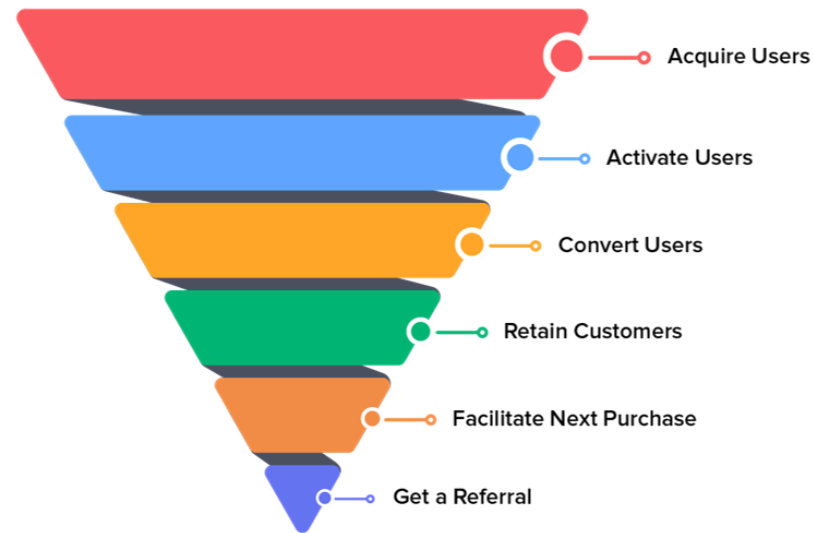
- Recommended actions and strategies based on champion/challenger testing strategy outcomes
- Applying advanced analytic techniques to make specific recommendations

Source: Principa at www.principa.co.za

9

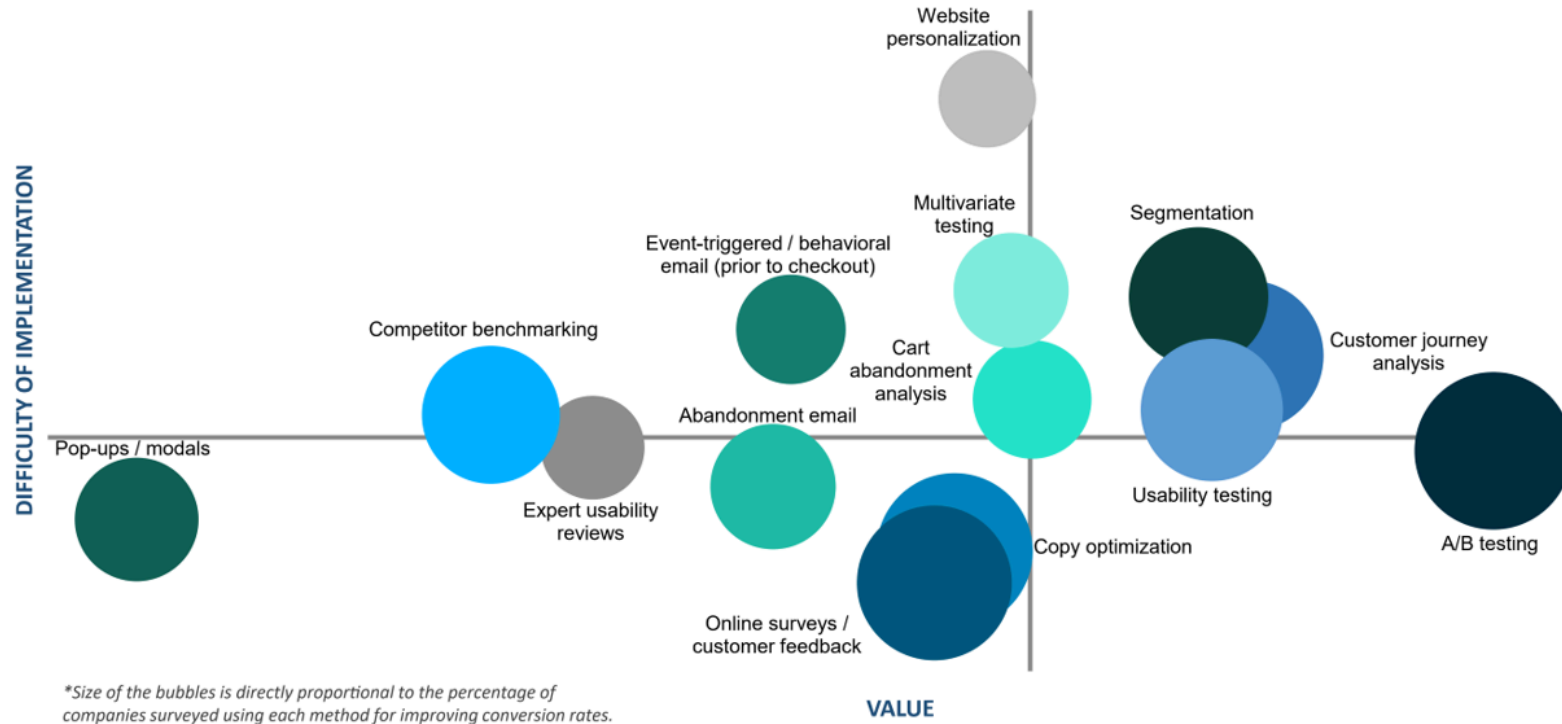
Descriptive (what happened), diagnostic (why), predictive (what will happen), and prescriptive (what should we do). Which type is hardest?

E-commerce Funnel



Retail has always been an early adopter of customer analytics. E-commerce funnels illustrate descriptive and diagnostic analytics in action. What causes drop-offs at each level?

E-commerce Analytics

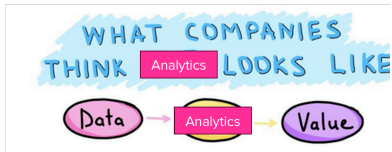


800 e-commerce pros were surveyed; companies using 9+ data-driven methods were most satisfied with their conversion rates. The optimal number of A/B tests was 3-5 per month.

Signs of a great analytics org

- C-level champion(s), i.e. C{D,E,A,F,M,O}O
- Centralized team regulates data, arch., standards & tools
- Decentralized analysts collaborate with execs
- Analytics career tracks are well established
- Careful in-housing/outourcing decisions about analytics
- Good examples?

Why is analytics hard?



- **Executives** may be territorial, prefer hunches, or misunderstand what data can and cannot do
- **Analysts** are expensive, hard to find, and not always interested in the business context
- **Culture** determines whether analytics makes decisions or merely justifies them; low-trust environments punish messengers and underuse data

Analytics works best when leadership creates an environment that enables investments in analytical frameworks and rewards disciplined decisionmaking, with retrospective decision evaluations and continuous-learning feedback loops. How have you seen analytics used in practice?

Analytics truisms

- Analytics matters more in B2C than B2B (why?)
- Selection effects are usually large
 - treatment effects are usually small
 - Key exceptions: Price or “free” giveaways
- Demographics don’t predict behavior very well
- Agencies lie about data sometimes
- “If it’s written in LaTeX, it’s probably correct”
- Stable analytics->decision loops should be automated and monitored carefully

Vocab

Common language facilitates communication

Customer level

- Core need: identifiable problem a customer wants to solve. Could be functional, emotional, social, profit-motivated, etc. Related: desire, want, pain point
- Core benefit: Customer's desired outcome of a purchase. E.g., commuters need to get to school, not necessarily cars
- Consumer: Entity that experiences the core benefit
- Customer: Entity that purchases and pays

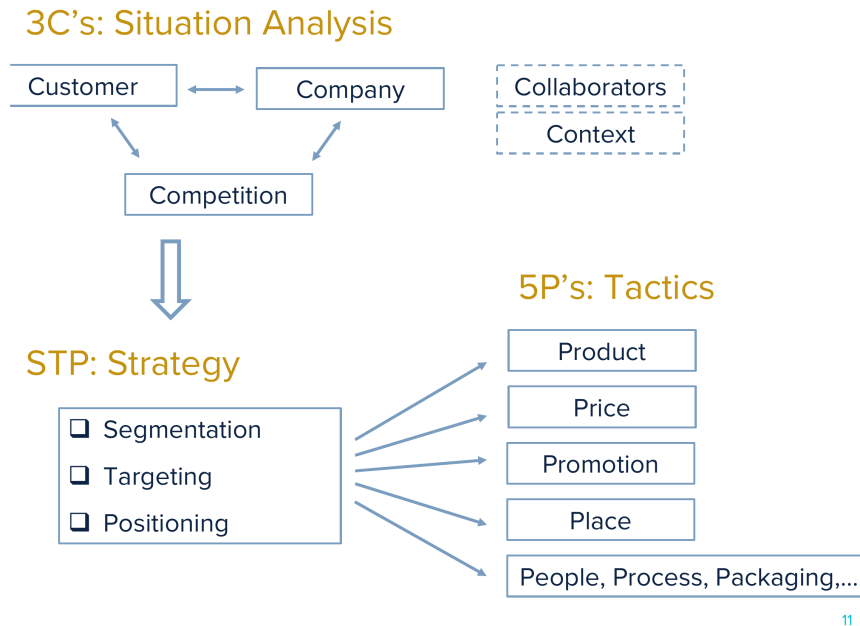
Product level

- Product/service/experience: Distinct offering that provides the core benefit
- Features: Aspects of a product that provide additional tangible or intangible benefits
- Value proposition: $\text{utility}(\text{Core benefit} + \text{features} - \text{price})$
- Contribution margin: $\text{Price} - \text{marginal cost}$
- Competitor: Any paid or free alternative that addresses the core need.
E.g., commute by bike, walk, bus, trolley, Uber, scooter, skateboard; work from home

Market level

- Market: Potential customer group with common core need
- Segment: Distinct subgroup of similar customers
- Targeting: Which segment(s) a firm tries to serve
- Positioning: Specification of product features to suit targeted segments
- Marketing: Practice of meeting customer needs profitably
 - Marketing: Business discipline that focuses most on customers
 - Ads & sales: Worthless without good value prop and positive margin
 - Poor implementation and short-term focus commonly lead to confusion with bullshit (“persuasive speech without regard for the truth” –Frankfurt 2005)

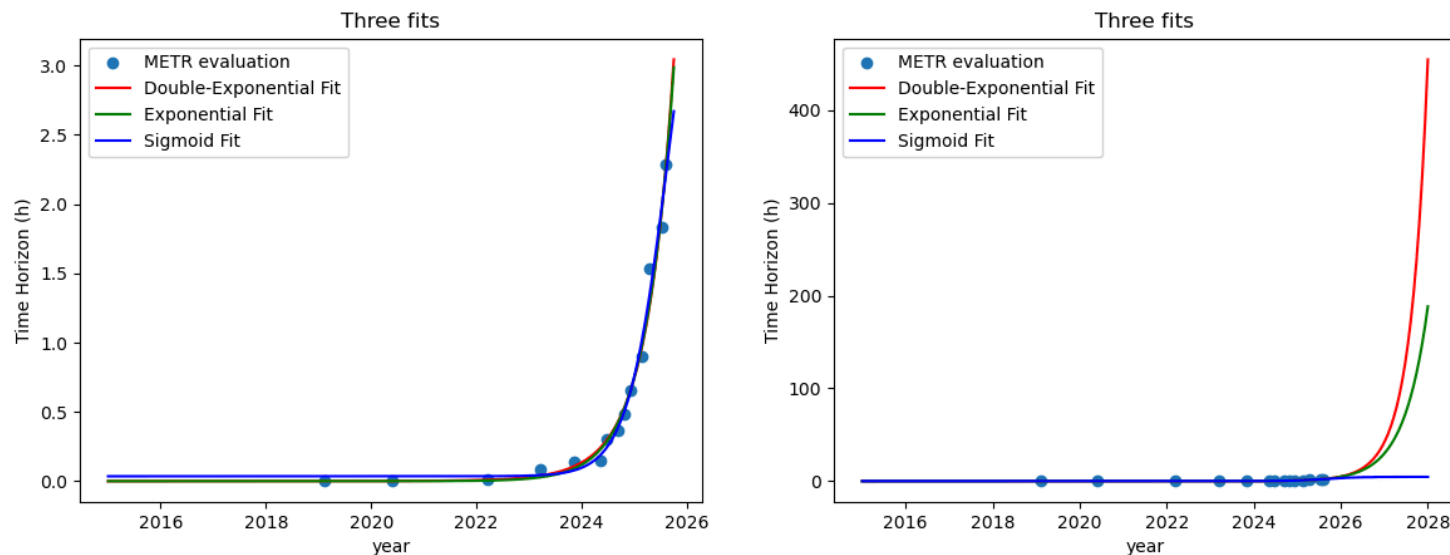
Legacy Terms



You need to know these well if you interview for marketing roles. Generations of marketing professionals were educated to think this way. Still relevant, but less, thanks to customer data abundance & analytics.

MGT 100 Principles

- Survey the field broadly, pointers for deeper learning. [Why R?](#)
- Piazza for all asynchronous interaction. No email or canvas messages
 - Important for fairness, timeliness, contribution scores & instructional team collaboration
- “When it comes to LLMs, skillful prompting leaves amateurs in the dust.”



Y axes indicate the human-expert task-completion time (in hours) that a frontier agent can complete with 50% reliability. Cox (2006) said “How [the] translation from subject-matter problem to statistical model is done is often the most critical part of an analysis.” What is model mis-specification? How does this figure into the public conversation about future AI capabilities? Will AI become all-powerful or should you study and build skills?



Please write down your intentions for this class. How will you measure your effort? Please don't say grades alone; grades are outcomes, not inputs.

Coding & Script

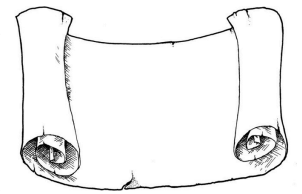
- All LLMs continue to hallucinate and require hand-holding. Beware free LLMs, they can be worse than useless.
- Conjecture:
(debugging difficulty) is exponential in (lines of code)
- We can code fast or slow. “Go slow to go fast”
- Pipe: $y \leftarrow f(g(x))$ is the same as $y \leftarrow x \mid> g \mid> f$
 - Old pipe was `%>%` ; remains widely used



Bad habit: Write the whole script, run it, see where it breaks. Good habit: test each chunk before writing the next one.

Today's script

- Data Import/export
- Data manipulation, summarization
- 5 verbs: Summarize, select, filter, arrange, mutate, group_by
- Univariate statistics
- Univariate plots
- Bivariate statistics
- Bivariate plots



Wrapping up

Competition

Week (relative to endorsement)	Region	Goya sales
-4	Right-leaning	87
-3	Right-leaning	85
-2	Right-leaning	87
-1	Right-leaning	90
0	Right-leaning	140
1	Right-leaning	133
2	Right-leaning	110
3	Right-leaning	95
-4	Left-leaning	158
-3	Left-leaning	159
-2	Left-leaning	158
-1	Left-leaning	159
0	Left-leaning	176
1	Left-leaning	170
2	Left-leaning	162
3	Left-leaning	158

Visualize Goya's weekly average sales in right- and left-leaning regions pre/post a major endorsement.

```
Data: goyadata <- readRDS(url("https://raw.githubusercontent.com/kennethcwilbur/mgt100/main/data/goya_sales.rds"))
```

Your visualization should make a clear point and be very easy to understand.



Recap

- Customer analytics :
Using customer data to improve decisions
- Data Viz is the first step in customer analytics
- Marketing : Meeting customer needs profitably
- Analytics types:
Descriptive, Diagnostic, Predictive, Prescriptive
- Summarize, select, filter, arrange, mutate, group_by

Going further

- Data-enabled storytelling (Wilke 2019)
- GGplot2 YT video
- *R for Data Science (3e)*
- **Big Book of R**: Lovingly curated, well organized, free resource directory for nearly any R problem

